

Amirali Bavafa

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EDUCATION

Simon Fraser University

Bachelor of Science in Computer Science

Burnaby, BC

Exp. Grad. Date: Apr. 2027

- Relevant Coursework: Data Structures & Algorithms, FullStack Development, Database Systems, Software Engineering

EXPERIENCE

Full-Stack Developer Intern

Sep 2024 – April 2025

Nexuswave Inc.

Vancouver, BC

- Developed and deployed web applications using React, Express, and Node.js to handle enterprise management systems.
- Connected the server-side to the PostgreSQL to handle (queries) create, read, update, and delete action.
- Integrated cloud services (AWS S3 & EC2) for secure data storage and scalable deployments.
- Built web app platforms handling user authentication through APIs, as well as submissions, and reporting.
- Implemented unit and integration tests using Jest to ensure high-quality, and increasing code coverage to 90%.
- Collaborated in an Agile team, participating in sprint planning, backlog grooming, and stand-ups to deliver features efficiently.
- Built CI/CD pipelines using GitHub Actions with automated testing and deployment for staging and production.
- Continuously learned new technologies including Next.js, Node.js, and AWS to introduce innovations using AWS Lambda.

Research Assistant - Artificial Intelligence (Medical Data)

Jan 2024 – Jul 2024

Simon Fraser University

Burnaby, BC

- Constructed machine learning models in Python & C++ to improve kidney disease diagnostic on LifeLabs datasets.
- Implemented search algorithms (UCS, BFS) to optimize classification pipelines, reducing runtime by 30%.
- Diagnosed and resolved data inconsistencies and algorithmic inefficiencies, improving model reliability and runtime stability.
- Collaborated in an Agile research workflow, improving communication and documentation across cross-functional teams.
- Co-authored technical reports and collaborated with healthcare researchers to interpret results for publication.

PROJECTS

Emral | *Node.js, Express, PostgreSQL, React, REST API* | GitHub

May – Oct 2024

- Engineered a contest management web app using Node.JS, React, JQuery, and Postgresql to manage contest and user data.
- Built a live leaderboard using React Hooks for real-time score updates and seamless user experience.
- Designed modular backend services following microservices principles to improve scalability and maintainability.
- Leveraged AWS EC2 to deploy and run the Node.js application for server-side operations.
- Refactored backend modules to PostgreSQL, redesigning relational schemas and queries to improve data consistency.
- Exhibited initiative by designing and deploying new production features, implementing automated CI/CD pipelines.
- Implemented unique QR code generation via REST APIs to track participant scores in real-time, enhancing contest integrity.
- Built a secured, authenticated admin portal using Passport.js to manage contests, users, and reports, ensuring proper access control and data security.

Racoon Invasion | *Java, JavaSwing, Maven, Junit, Agile* | GitHub

Feb – Apr 2025

- Designed an arcade-style defense game with AI-powered enemies and collision mechanics, with 200+ downloads.
- Participated in Agile sprint planning, backlog refinement, and iterative development to implement gameplay features.
- Manufactured robust Java backend logic with emphasis on modular, maintainable code and unit testing, aligning with professional software development standards.
- Implemented rigorous JUnit test suite achieving 95% gameplay logic, and method coverage.
- Integrated frontend (React) and backend APIs for real-time leaderboard updates.
- Applied OOP principles and continuous testing for gameplay mechanics like collision detection and scoring.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, C/C++, SQL, Python, HTML/CSS

Frameworks & Libraries: Node.js, React, Next.js, Jest, Angular, JUnit, Bootstrap

Databases & Infrastructure: REST API, HTTP, JSON, PostgreSQL, MongoDB, AWS

Developer Tools: Git, Docker, Google Cloud Platform, Firebase, Postman